

CHANDRA SHEKHAR PANDEY

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SUMMARY

ML systems engineer specializing in large-scale MoE training, LLM inference, and low-precision GPU kernels. Delivered MXFP8 training and serving optimizations across AMD CDNA3/CDNA4 systems, scaled DeepSeek-V3 to 1,024 GPUs, and contributed performance improvements to PyTorch, TorchAO, and vLLM.

TECHNICAL SKILLS

Frameworks: PyTorch, TorchAO, TorchTitan, Megatron-LM, Primus-Turbo, TensorFlow, JAX

Distributed Training: Expert Parallelism, FSDP, RCCL/NCCL collectives, Slurm, Docker, Kubernetes

Inference and Post-training: vLLM, SGLang, Atom, vime, Miles, VeRL; GRPO, colocated rollout and weight synchronization

GPU and Kernels: Python, C/C++, CUDA, ROCm/HIP, Triton, Gluon, FlyDSL; rocprof, Kineto, occupancy tuning

Hardware and Precision: AMD CDNA3/CDNA4; NVIDIA Hopper/Blackwell; FP8, MXFP8, MXFP4, NVFP4

PROFESSIONAL EXPERIENCE

Member of Technical Staff — AMD, San Jose, CA

January 2025 – Present

Promoted from Senior Software Development Engineer; focused on distributed training, LLM inference, and GPU kernel performance.

- **MXFP8 training for DeepSeek-V3:** delivered the full MoE training path in TorchTitan/TorchAO with a FlyDSL backend on MI355X, now achieving 1.2× BF16 end-to-end performance. Designed a ragged MXFP8 wgrad grouped GEMM reaching 2.27 PFLOP/s (45% of peak), 13× over the prior Triton kernel and up to 3× over BF16 hipBLASLt; being upstreamed to PyTorch Inductor.
- **Large-scale MoE pre-training:** scaled DeepSeek-V3 671B to 1,024 MI325X GPUs with 96% scaling efficiency through Expert Parallelism strategies and FP8 grouped GEMM kernels; partnered with AMD and Meta PyTorch teams on TorchTitan/Primus-Turbo integration and co-authored published results reporting 2.77× end-to-end training speedup.
- **MXFP8/MXFP4 kernel optimization:** built CDNA4 forward and dgrad grouped GEMMs using XOR-swizzled LDS layouts, ping-pong software pipelining, XCD swizzle for L2 locality, fused quantization epilogues, and Split-K for K-heavy shapes; eliminated token-dispatcher copies and per-step pad-strip transposes and gated the path with bit-exact operator parity tests.
- **RL post-training on ROCm:** led end-to-end enablement of vime on MI355X across Megatron training and vLLM rollout, validated the GRPO pipeline, upstreamed ROCm fixes, shipped a prebuilt container, and published the work on the vLLM blog.
- **Inference enablement and performance analysis:** delivered launch-day ROCm support for OpenAI gpt-oss-120B/20B on MI300X/MI355X; built TTFT, throughput, concurrency, and Kineto/Perfetto workflows for MI355X vs. B200 comparisons and MoE prefill/decode roofline studies that guided serving and kernel decisions.
- **Recognition:** earned the Next 5% Award presented by AMD's CEO, along with 2 Executive Spotlight and 6 Spotlight awards within the first year at AMD.

Machine Learning Intern — Bytez Inc., San Francisco, CA

February 2024 – May 2024

- Fine-tuned Code Llama 13B into a text-to-Cypher model for natural-language graph querying and built semantic search over a Neo4j graph of approximately 3 million research papers.

Software Development Engineer — Bosch Global Software Technologies, India

January 2021 – July 2022

- Built Python automation frameworks for end-to-end testing across 12+ peer groups, reducing functional test time by 80% through scripted failure-case simulation.

OPEN SOURCE PROJECTS

- **unet.cu:** built a UNet diffusion training and inference framework in C++/CUDA with HIP support, reaching 40% of PyTorch plus torch.compile end-to-end training performance.
- **llm.c:** ported CUDA kernels to HIP to add AMD GPU support to Andrej Karpathy's project.

TECHNICAL WRITING AND PUBLICATIONS

- **AMD ROCm Blogs:** authored six published posts covering MI355X occupancy, gpt-oss day-0 enablement, DeepSeek-V3 profiling, Llama 4 inference, AITER, and Gemma 3 deployment. rocm.blogs.amd.com/authors/shekhar-pandey
- **vLLM Blog:** "vime + ROCm: End-to-End RL Post-Training on AMD Instinct GPUs" (July 2026).
- **Co-authored publications:** AMD and Meta PyTorch results on TorchTitan/Primus-Turbo training speedups; reproducibility research presented as a poster at NeurIPS 2023.

EDUCATION

M.S. Computer Engineering, GPA 3.9/4.0 — New York University

August 2022 – May 2024

ADDITIONAL

Top 0.76% on LeetCode.